

EXECUTIVE BUSINESS REPORT

End-to-End Sales Forecasting & Demand Intelligence System

Prepared for: Head of Supply Chain & CFO | Aditi Joshi, Data Science Intern | July 2026

1. EXECUTIVE SUMMARY

This analysis was conducted using four years of Superstore sales data (2015–2018), covering 4,922 customer orders with total revenue of approximately \$2.26 million. Sales increased by around 50% during the study period. Technology generated the highest revenue (\$827K), while the West region contributed the largest share of sales (\$710K). Three forecasting approaches were evaluated, and based on the one with the lowest error rate, expected revenue for the next three months is approximately \$261K. The analysis also identified 17 unusual sales weeks and grouped all 17 product sub-categories into four demand clusters. Based on these findings, several actions are suggested to improve inventory planning and regional sales performance.

2. KEY FINDINGS FROM DATA ANALYSIS & FORECASTING

Revenue by Category			Revenue by Region		
Category	Revenue	Share	Region	Revenue	Share
Technology	\$827,456	37%	West	\$710,220	31%
Furniture	\$728,659	32%	East	\$669,519	30%
Office Supp.	\$705,422	31%	Central	\$492,647	22%
			South	\$389,151	17%

Revenue grew from \$479K in 2015 to \$722K in 2018. There was a small decline in 2016 (–4%) followed by strong recovery. The highest single month on record was November 2018 (\$117,938), which the analysis suggests was driven by holiday season demand.

3. THREE-MONTH SALES FORECAST

Three forecasting approaches were evaluated: SARIMA, Facebook Prophet, and XGBoost. XGBoost achieved the lowest forecasting error (MAPE 14.8%), performing substantially better than SARIMA (41.6%) and Prophet (40.0%). It was therefore selected as the model for forward-looking estimates. Based on current trends, projected revenue for the next three months is approximately \$261,000. Actual performance may vary depending on promotions, seasonality and market conditions.

Period	Projected Sales	Indicative Range	Note
Month 1	\$88,694	\$75,000 – \$102,000	Stable demand continuation
Month 2	\$86,699	\$73,000 – \$100,000	Stable demand continuation
Month 3	\$85,878	\$72,000 – \$100,000	Slight seasonal tapering

4. UNUSUAL SALES EVENTS (ANOMALY DETECTION)

All 209 weekly sales periods were scanned using two independent methods — Isolation Forest and Z-Score analysis. Both methods agreed on five events, which increases confidence in those findings. In total, 17 anomalies were identified: 12 upward spikes and 5 downward drops. The three largest spikes are shown below.

Rank	Date	Weekly Sales	Type	Likely Cause
#1	Mar 2015	\$37,704	Spike	Possible bulk order or unplanned promotion
#2	Nov 2018	\$35,999	Spike	Holiday season / Black Friday demand
#3	Nov 2018	\$30,572	Spike	Continued holiday season demand

5. PRODUCT DEMAND SEGMENTATION

All 17 sub-categories were grouped into four clusters based on their sales volume, growth rate, and revenue volatility. The clustering suggests that most revenue is concentrated in a small number of stable, high-volume products, while several sub-categories show little or no growth. Each cluster points to a different approach for inventory management.

Cluster	Sub-Categories	Revenue	Growth	Suggested Inventory Approach
High Volume Stable Demand	Accessories, Binders, Chairs, Phones, Storage, Tables	\$1,436,975	+25%	Keep stock levels high with automated reorder points. Avoid running out.
Growing Products	Appliances, Art, Labels, Paper	\$220,500	+44%	Increase order quantities gradually. Review performance each quarter.
Seasonal / Volatile	Copiers, Machines	\$335,487	+7%	Order in advance of Q4. Reduce stock outside peak periods.
Low Demand Products	Envelopes, Fasteners, Furnishings, Bookcases	\$268,576	~0%	Consider drop-shipping or lower stock levels to free warehouse space.

6. SUGGESTED ACTIONS

Based on these findings, three areas stand out as worth acting on to improve sales performance and reduce inventory risk.

① Increase sales efforts in the South region	The South contributed only 17% of total revenue (\$389K) compared with 31% from the West (\$710K). The data suggests there may be room to grow sales in this region through more targeted marketing or stronger sales support. Even a modest improvement in South region performance could add a meaningful amount to overall revenue.
② Prepare inventory before the fourth quarter	Historical data shows that November consistently sees higher sales. Two of the three largest anomalous weeks in the entire dataset occurred in November 2018, with weekly sales exceeding \$35,000. Ordering additional stock for Technology and Furniture ahead of Q4 would reduce the risk of running short during the busiest sales period.
③ Review low-performing product lines	Five sub-categories (Envelopes, Fasteners, Furnishings, Supplies, Bookcases) generated a combined \$268K over four years with near-zero growth. These products may be worth reviewing — lower stock levels or drop-shipping could free up warehouse space for faster-moving items without significantly impacting revenue.

7. LIMITATION

The forecasting models used in this analysis are based entirely on historical sales patterns. They do not account for external factors such as changes in competitor pricing, broader economic conditions, supply disruptions, or new product introductions. As a result, the projections should be treated as indicative estimates rather than firm predictions. The analysis would benefit from being updated periodically as new sales data becomes available.
